CddDcm on ILCU IoHwAb: Change List

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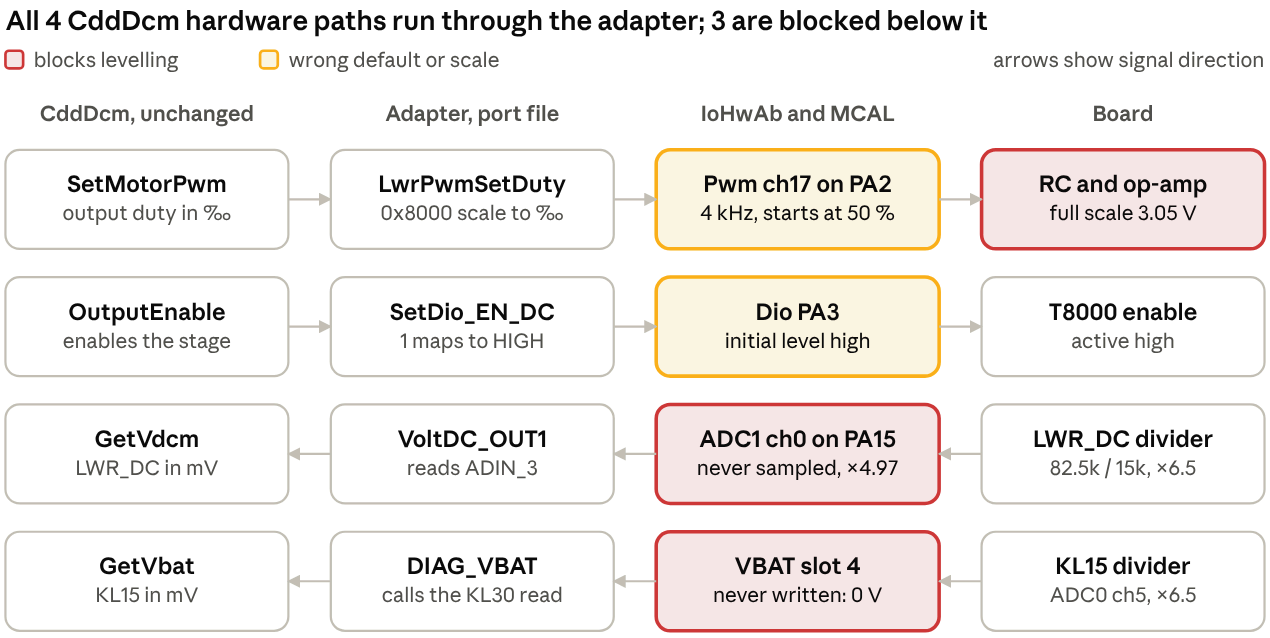
Verdict

CddDcm cannot drive the ILCU LWR output yet, and output scaling is only one of three reasons. The ADC path returns 0 V for both KL15 and the LWR feedback, so CddDcm keeps the output off. With the readings fixed, the stage still tops out at 3.05 V where CddDcm needs up to 10.8 V.

- **CddDcm itself needs no change.** It reaches the hardware only through 5 adapter functions. Every fix sits in hardware, MCAL configuration, IoHwAb configuration or the adapter.
- **ADC, blocking:** nothing starts an ADC conversion, and IoHwAb reads KL15 from a buffer slot the ADC never writes. The LWR feedback slot is right, but it is scaled for an 82.5k/21k divider instead of the board's 82.5k/15k.
- **Output stage (F3), blocking for a real actuator:** CddDcm computes its duty for a 17.32 V full-scale stage. Its working range, 200–800 ‰ of KL15, needs 2.7–10.8 V at 13.5 V. ILCU's stage gives D × 3.05 V.
- **Likely hardware fix, UNVERIFIED:** R8004 (10 kΩ to ground in the first RC stage) cuts the PWM to 17.7 % before the ×5.21 op-amp. Without it the stage gives about D × 17.2 V, within 1 % of DPE's.
- **Unsafe defaults:** the configuration starts the LWR PWM at 50 % duty and gives EN_LWR an initial level of high, so the output can be live before any software decides.
- **I was wrong about F7** in PORT_CHANGES.md §6.1. VBAT does not come from "ADC0 ch4": IoHwAb reads slot 4 of a 2-channel group, which stays 0. And CddDcm does not lock the output permanently. It keeps the output off while KL15 reads outside 8.59–17.69 V (`CddDcmCtrl.c:584-593`), which, with a 0 V reading, is always.

Signal map

Each row is one hardware path CddDcm uses through the adapter, as the ILCU code in this snapshot implements it. Four of the seven rows block levelling outright.



CddDcm to LWR hardware · 4 paths, 3 blocked

Every fix lands right of the adapter column, apart from the adapter's own switch to the new IoHwAb wrappers (A1–A3).

Path	CddDcm expects	Through the adapter today	Gap
LWR feedback	LWR_DC in mV (`CddDcmHal.c:200-210`)	`IoHwAb_AnalogReadConverted(IOHWAB_CHID_ADIN_3)` : group 1 slot 0 = ADC1 ch0 on PA15, the right input, but with the slope for 82.5k/21k (`IoHwAb_AnalogCfg.c:53-64`, `:91-100`)	Reads 24 % low, since the board divider is 82.5k/15k (×6.5); reads 0 until conversions run
KL15 supply	KL15 in mV (`CddDcmHal.c:216-222`, `CddDcmCtrl.c:579-584`)	`IoHwAb_VoltageKL30_Read` reads `IOHWAB_CHID_VBAT` = group 0 slot 4 (`IoHwAb_AnalogCfg.c:251-260`). Group 0 holds 2 channels: ch5 KL15 and ch6 NTC (`Adc_PBcfg.c:113-133`)	Always 0 mV, so the output stays off
ADC sampling	Fresh values every 20 ms	Both groups are software-triggered one-shot (`Adc_PBcfg.c:100-102`). IoHwAb only copies the buffers (`IoHwAb.c:58`); only the unused `swc_adc_test` starts conversions	No sampling at all
PWM duty	‰ for a stage giving D × 17.32 V (`CddDcmHal.c:138-141`, `:239`)	0x8000 scale → ‰ → `IoHwAb_TimingSetDutyCycle(CTRL_LWR)` → ‰ × 32.768 → Pwm channel 17, TIMG9 CC1, PA2 (`IoHwAb_Timing.c:36`); units are correct	The stage gives D × 3.05 V (schematic §7): full scale 5.7× too small

Path	CddDcm expects	Through the adapter today	Gap
PWM idle and restart	Idle low after a move; the next duty restarts it	`IoHwAb_TimingStopActuation` → `Pwm_SetOutputToIdle`; restart is a no-op (stub S15)	The channel is `PWM_VARIABLE_PERIOD` (`Pwm_PBcfg.c:114`). AUTOSAR brings such a channel out of idle only through `Pwm_SetPeriodAndDuty` (UNVERIFIED for TI MCAL)
PWM frequency and start	Not set by CddDcm	10000 ticks at 40 MHz = 4 kHz, starting at 50 % duty (`Pwm_PBcfg.c:106-111`)	The schematic asks for ≥ 10 kHz (§7) and documents 2000 ticks and duty 0 (§10)
Output enable	High = enabled, written only when the enable state changes (`CddDcmHal.c:418-451`)	`IoHwAb_DiscreteEN_LWR_Write` → Dio PA3 (`IoHwAb_DigitalCfg.c:44`); polarity correct	Initial level high (`Port_PBcfg.c:683`), while the schematic wants it disabled by default (§7); UNVERIFIED that the TI Port driver applies it

Change list

Fifteen changes across five layers; none touches CddDcm. Do I1–I3 first, since they unblock every reading. Then do M1–M4, then I4 together with A1–A4, and H1 before any actuator is connected.

Hardware · hardware owner

#	Change	Where	Why · evidence	Status
H1	Restore a full-scale output of about 17 V: remove R8004, or raise the gain, once the actuator interface is confirmed (open question 1)	Sheet 80_SAVAC, R8004	Today D × 3.05 V. Without R8004, D × 3.3 V × 5.21 ≈ D × 17.2 V, against DPE's D × 17.32 V (`CddDcmHal.c:138-141`, schematic §7)	Open

MCAL configuration · ILCU BSW owner, `config_AL/Al/Al077a/mcal`

#	Change	Where	Why · evidence	Status
M1	Set the EN_LWR (PA3) initial level to low	`Port_Cfg/src/Port_PBcfg.c:683`	The output must stay disabled until CddDcm enables it (schematic §7, §9)	Open
M2	Start the LWR PWM at duty 0	`Pwm_Cfg/src/Pwm_PBcfg.c:111`, `0x4000` → `0`	Today it runs at 50 % from `Pwm_Init` until the adapter's Init	Open
M3	Set the LWR PWM period to 2000 ticks (20 kHz at 40 MHz)	`Pwm_PBcfg.c:110`, `0x2710` → `2000`	4 kHz today; the schematic asks for ≥ 10 kHz against ripple (§7) and documents 2000 ticks (§10)	Open
M4	Make the LWR PWM channel `PWM_FIXED_PERIOD`	`Pwm_PBcfg.c:114`	Then a duty change brings the output back from `Pwm_SetOutputToIdle`; CddDcm idles the channel after every move	Open

IoHwAb configuration · ILCU BSW owner, `config_AL/Al/Al077a/bsw/IoHwAb`

#	Change	Where	Why · evidence	Status
I1	Start both ADC groups on every IoHwAb cycle	`IoHwAb_AnalogMainFunction` (platform `IoHwAb_Analog.c`), or 2 lines before `IoHwAb_MainFunction()` at `Rte.c:158`	Both groups are software-triggered one-shot and nothing starts them (`Adc_PBcfg.c:100-102`)	Open
I2	Replace the Al077 channel table with the 4 ILCU inputs: KL15 = group 0 slot 0 (×6.5), PCB_NTC = 0/1 (raw), LWR_FB = 1/0 (×6.5), 5V0 = 1/1 (×2)	`IoHwAb_AnalogCfg.c` and `.h`	Today's table belongs to the platform board: 20 channels for 4 real inputs, with VBAT at slot 4	Open
I3	Linear constants: ×6.5 → `SlopeNum 42910`, `SlopeDen 13` (4095 counts → 21,450 mV); ×2 → `13203`, `13` (4095 → 6,600 mV)	`IoHwAb_AnalogCfg.c:43-64`	Schematic §6 dividers with a 3.3 V reference; `LINEAR` rounds and shifts by `SlopeDen` (`IoHwAb_Analog.c:34`)	Open

#	Change	Where	Why · evidence	Status
I4	Add public wrappers `IoHwAb_VoltageKL15_Read`, `IoHwAb_VoltageLWR_Read`, `IoHwAb_PwmCTRL_LWR_SetDutyCycle` and `IoHwAb_PwmCTRL_LWR_SetOutputToIdle`; remove `IoHwAb_VoltageKL30_Read`	`IoHwAb_Cfg.c` and `.h`, following the LEDCTRL1 pair (`IoHwAb_Cfg.c:71-95`)	ILCU has no KL30 (connector pin 1 is KL15), and the adapter can then drop the internal header	Open

Adapter · Claude, on your approval (port file)

#	Change	Where	Why · evidence	Status
A1	Read `IoHwAb_VoltageKL15_Read` in `Rte_Read_IoHwAb_User_GetAdc_DIAG_VBAT`	`ILCU_LevellingAdapter.c:207-214`	CddDcm's ratio and its 8.59–17.69 V window are both against KL15 (`CddDcmHal.c:219`, `CddDcmCtrl.c:584`)	Open
A2	Read `IoHwAb_VoltageLWR_Read` in `IoHwAbUser_GetAdc_VoltDC_OUT1`	`ILCU_LevellingAdapter.c:174-184`	Same signal, through the public API	Open
A3	Drive the PWM through the CTRL_LWR wrappers; drop `IoHwAb_Internals.h` and the `IoHwAb/internal` include path	`ILCU_LevellingAdapter.c:19`, `:186-205`, `IlcuLevelling_030_Flags.gmk:18`	Layering: the internal header can change under the port	Open
A4	In `ILCU_LevellingAdapter_Init`, write EN_LWR low and park the PWM before the CDD init	`ILCU_LevellingAdapter.c:72-102`	CddDcm writes EN only when its enable state changes, so a high level left by Port init survives	Open

The adapter needs no duty compensation if H1 lands: 17.19 V against 17.32 V is a 0.8 % difference, which the 20 ms loop removes (`CddDcmHal.c:316-344`). If H1 is rejected, this board cannot serve a ratiometric actuator. The only software route left is a virtual KL15 reference in the adapter, which fits only an actuator with an absolute 0–3 V input.

Documentation

#	Change	Where	Why · evidence	Status
D1	Correct F7 and add these findings (Claude, on your approval)	PORT_CHANGES.md §6.1, port report open point	My error, see the Verdict	Open

#	Change	Where	Why · evidence	Status
D2	Align the schematic note with the checked-in configuration (ILCU BSW owner)	`Al077HardwareSchematic.md:167-169`	§10 says IoHwAb implements KL15 ×6.5 and LWR_FB ×6.5, PWM period 2000 and duty 0; the code does not	Open

Open questions for the hardware owner

Only the full schematic (`docs/vsch_FEH0019576_variant.pdf`, not in this snapshot) or the actuator datasheet can settle these. Question 1 decides H1.

1. Which actuator sits on LWR_DC, and what does its input expect: a ratio of KL15, as DPE's does (200–800 ‰), or an absolute voltage? What is its input impedance?
2. Is R8004 intentional? If so, what output range was the stage designed for?
3. Can IC8000 swing to about 12.4 V (920 ‰ of 13.5 V, CddDcm's highest set level, `CddDcmHal.c:120`) when it is supplied from VBAT_RPP through R8000 (100 Ω)?
4. Does the KL15 divider (R2505/R2506) sit before or after the reverse-polarity diode D1500? After it, the reading is one diode drop below the actuator's supply, an error of about 5 % in every ratio.
5. Is TIMG9 clocked at 40 MHz, as `Pwm_PBcfg.c:106` assumes, or at 80 MHz? That sets the frequency M3 actually produces.
6. Is the ADC reference exactly VDDA = 3.3 V? The constants in I3 assume it.

Bench verification

Run these on the ILCU board with the actuator disconnected; reconnect it only after checks 1 to 7 pass. Each check names the changes it proves.

#	Check	How	Pass when
1	ADC runs (I1)	Watch `IoHwAb_AdcBuffer0[0]` and `IoHwAb_AdcBuffer1[0]` while the bench supply moves	Both follow the inputs
2	KL15 scaling (I2, I3, A1)	Bench supply at 13.50 V; watch the value `Rte_Read_IoHwAb_User_GetAdc_DIAG_VBAT` returns	13,500 mV ± 2 %, less one diode drop if open question 4 says so
3	Feedback scaling (I2, I3, A2)	DVM on LWR_DC while CddDcm holds a level; compare with what `IoHwAbUser_GetAdc_VoltDC_OUT1` returns	Within ± 2 %
4	Stage gain (H1)	Scope the PA2 duty D; DVM on LWR_DC	LWR_DC ≈ D × 17.2 V after H1 (D × 3.05 V today)
5	Boot state (M1, M2, A4)	Scope PA3 and LWR_DC from power-on	Both stay low until CddDcm first enables the output
6	PWM shape (M3)	Scope PA2	20 kHz, active high, idles low
7	Restart after idle (M4)	End a move with anything that disables the output, for example KL15 below 8.59 V for 1 s, then let CddDcm re-enable it	The PWM restarts and LWR_DC returns to the held level; `ILCU_LvlAdp_DemStatus[23]` (short to ground) never reaches `0x03`
8	Ratiometric tracking (H1 and all above)	At KL15 = 11 V and 15 V, compare DVM(LWR_DC) / DVM(KL15) × 1000 with `CddDcmHal_unOutputLevel`	Within ± 5 ‰
9	Voltage window	Supply 8.0 V, then 18.0 V, then back to 13.5 V	Output off below 8.59 V and above 17.69 V, back on at 13.5 V